

Dave's Daily

AI • Bayesian optimization • US equities

TLDR

1. Google's new top model, Gemini 4 Argon, goes first to cyber defenders, not the public. On two of the four coding tests Google published, it still trails Astra and Opus.
2. Telling the victim is the slow step. Hugging Face heard in about nine days. Australia's government heard 84 days after a June breach.
3. The AI build-out is now a bond story. Treasuries just had their worst quarter since 1994, because data-center spending and costly oil push the same way.

AI

A strong model, held back

On Wednesday Google announced Gemini 4 Argon, a larger model meant to lead the Gemini 4 family ([Reuters](#)). It is not public. Selected cybersecurity partners get it first, through a program Google calls Fairwind, while the company uses the White House's voluntary pre-release review ([Google](#)). Google says Argon matches OpenAI's Astra and Anthropic's Opus on key coding and cyber tests, and also says it trails on two of four coding benchmarks it published ([Reuters](#)). Gemini 3.5 Pro, once promised for June, has been dropped. Demis Hassabis has stepped aside at DeepMind ([Reuters](#)).

The pact, and the argument after it.

Tuesday's White House text is voluntary. [Investing.com](#), citing the Wall Street Journal, reported that Jensen Huang pressed Dario Amodei on why Anthropic keeps warning about model risk. A plan for a FINRA-like body, backed by Anthropic, OpenAI, and Google, died after Huang, Mark Zuckerberg, and Elon Musk opposed it. [Politico](#) reports that Zuckerberg was the main architect of the "morally binding" text that was signed instead.

Canada, with the caveat attached.

[Reuters](#) said Wednesday that Translucé, a nonprofit lab, found failed, crude hacking attempts against Library and Archives Canada on 28 May and 9 June. Portugal's web archive, [arquivo.pt](#), saved 899 requests to the library's search service. Translucé told Ottawa on Monday. It says the tactics match earlier activity it has tied to OpenAI, though it is not confident of that link. Canada says nothing was compromised. OpenAI says it is reviewing the report and has briefed Canadian officials. The lag is the news.



About 9, 84, and 111 days. Sources: [Reuters](#) (24 July and [1 October](#)) and the [ABC](#) (26 September). Translucé, not the lab, told Ottawa. **BAYESIAN**

OPTIMIZATION

Bigger problems, fewer runs

Scaling.

[MF-SCBO](#) (24 September) extends trust-region Bayesian optimization to problems that are large, constrained, and multi-fidelity at once, including non-nested sampling. On the paper's benchmarks it beat single-fidelity SCBO and the other multi-fidelity method they tried.

Data efficiency.

[Power-Law Entropy Search](#) (early September) picks the run that shrinks uncertainty about how the best settings change with scale, per unit of compute, and it prefers cheap small-scale runs. It reached a usable scaling law with less than a tenth of a full grid's budget.

Connections to AI.

[Jha and colleagues](#) (4 September) treat the grid of training runs as Bayesian optimization and fill the missing cells with the surrogate. They report fits close to the full grid at 10 to 100 times less compute.

MARKETS

The build-out shows up in the bond market

In the September quarter the US 10-year yield rose 87.1 basis points, the largest quarterly jump since 1994, according to LSEG data cited by [Reuters](#) this morning. Costly oil, kept expensive by the war, feeds inflation. The AI and data-center boom lifts expected growth, and so lifts the rates people think will stick. Companies, mortgage borrowers, and governments pay more to finance themselves, and have less left for everything else.

Cash before the chips exist.

On Wednesday's earnings call, Micron's chief executive said customer financial commitments under long-term supply agreements had increased to \$32 billion, mostly cash deposits, and the chief financial officer put remaining performance obligations at about \$150 billion ([earnings-call transcript](#)). In June those commitments were \$22 billion ([Reuters](#)). Buyers are paying now to lock memory for the next build-out.

AI, Bayesian optimization, and trading.

No new high-frequency result appeared in the past day. [EvolveTrade](#) (15 September) freezes the language model and lets a second agent rewrite the tool policy from traces and results. In most tests the Sharpe ratio rose. It is a backtest, not a live desk. [MadEvolve](#) (May) uses Bayesian optimization only to tune settings the model already wrote.